

Mini HCM Time Tracking - Activity

Overview

A light-weight Human Capital Management (HCM) Time-In/Time-Out system built with free tools: **Firebase (free tier for Auth + Firestore)** for data/auth, **React.js** (open-source) for the frontend, and **Node.js (Express)** (open-source) for server-side processing. The system records employee punches for for a **1-week duration task**, computes worked hours, overtime (OT), night differential (ND), lateness, undertime, and keeps aggregated daily summaries.

Activity

The activity is to create a basic HCM system with the following features:

1. User Registration & Authentication

2. Use Firebase Authentication (Email/Password).

3. Store user details (name, email, role, timezone) in Firestore. Each user document should also include a **schedule field** that defines their shift start and end times (e.g., `schedule: { start: '09:00', end: '18:00' }`). This schedule is then used for computing late arrivals, undertime, and overtime.

4. Time-In / Time-Out Logging

5. Build a React interface with **Punch In** and **Punch Out** buttons.

6. Save punches to Firestore under an `attendance` collection.

7. Each punch includes timestamp and user ID.

8. Computation of Hours

9. Implement logic (Node.js/Express or Firebase Functions) to calculate:

- **Regular hours** (up to scheduled shift).
- **Overtime (OT)** (beyond shift).
- **Night Differential (ND)** (work between 22:00–06:00).
- **Late** (arrival after shift start).
- **Undertime** (leaving before shift end).

10. Daily Summary

11. Store daily totals in a `dailySummary` collection.
 12. Display results in React (Dashboard + History table).
 13. Include breakdown of regular hours, OT, ND, late, and undertime.
 14. **Admin Tools**
 15. Admin role can view/edit punches.
 16. Admin can view daily reports of employees with all metrics (regular, OT, ND, late, undertime).**
 17. Admin role can view/edit punches.
 18. Admin can view weekly reports of employees with all metrics (regular, OT, ND, late, undertime).
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Tech Stack

- Frontend: React.js (open-source)
 - Backend: Node.js + Express (open-source)
 - Database/Auth: Firebase (Firestore + Firebase Auth) — **Free tier**
 - Hosting: Firebase Hosting (free tier) for React, optional free tiers for backend (Vercel/Render/Heroku)
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Expected Output

- A working mini HCM system where an applicant demonstrates:
 - Registration and login flow.
 - Time-in/time-out recording.
 - Automatic computation of regular hours, overtime, night differential, lateness, and undertime.
- A Daily summary dashboard with full breakdown. The daily dashboard should display metrics such as daily regular hours, overtime, night differential hours, late arrivals, and undertime, with key performance indicators (KPIs) for quick monitoring.